

A retail company receives daily order files from its sales application.

The source application uploads CSV files into an **Azure Storage Account**.

Snowflake should automatically load the new files whenever they arrive in Azure Storage.

The company wants:

1. A separate Snowflake warehouse for loading data
 2. A separate user and role for data engineering
 3. Data ingestion from Azure Storage Account
 4. Initial load from CSV file
 5. Automatic incremental loading when new CSV files are uploaded
 6. Avoid duplicate records in the final table
-

Important Concept for Students

Snowflake **does not automatically detect new rows added inside the same old file**.

Best practice is:

Do not edit the already uploaded CSV file.

Instead, upload a new CSV file for every new batch.

Example:

retail_orders_initial.csv

retail_orders_incremental_20260508.csv

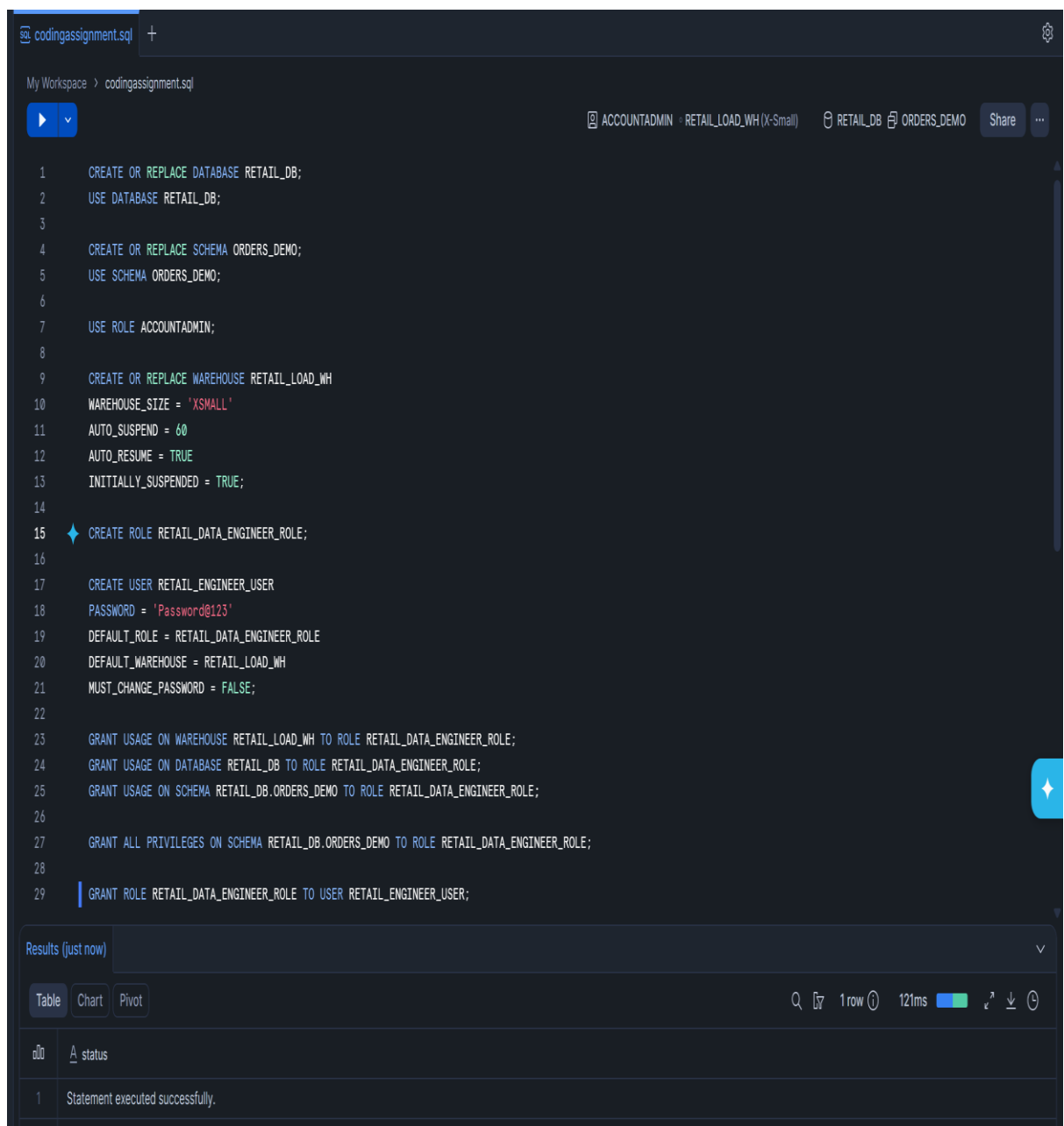
retail_orders_incremental_20260509.csv

retail_orders_incremental_20260510.csv

Snowpipe detects and loads **new files**, not newly edited rows inside the same file.

A: Snowflake Setup

1. Create a database named RETAIL_DB.
2. Create a schema named RETAIL_DB.ORDERS_DEMO.
3. Create a warehouse named RETAIL_LOAD_WH.
4. Create a role named RETAIL_DATA_ENGINEER_ROLE.
5. Create a user named RETAIL_ENGINEER_USER.
6. Grant required permissions to the role and user.



The screenshot displays a Snowflake SQL editor interface. The top bar shows the file name 'codingassignment.sql' and a '+' icon. Below the bar, the workspace path is 'My Workspace > codingassignment.sql'. The main editor area contains a SQL script with 29 lines of code. The script includes commands to create a database, a schema, a warehouse, a role, a user, and grant various permissions. The bottom panel shows the 'Results (just now)' section with a 'Table' tab selected. The table has one row indicating that the statement was executed successfully.

```
1  CREATE OR REPLACE DATABASE RETAIL_DB;
2  USE DATABASE RETAIL_DB;
3
4  CREATE OR REPLACE SCHEMA ORDERS_DEMO;
5  USE SCHEMA ORDERS_DEMO;
6
7  USE ROLE ACCOUNTADMIN;
8
9  CREATE OR REPLACE WAREHOUSE RETAIL_LOAD_WH
10 WAREHOUSE_SIZE = 'XSMALL'
11 AUTO_SUSPEND = 60
12 AUTO_RESUME = TRUE
13 INITIALLY_SUSPENDED = TRUE;
14
15 CREATE ROLE RETAIL_DATA_ENGINEER_ROLE;
16
17 CREATE USER RETAIL_ENGINEER_USER
18 PASSWORD = 'Password@123'
19 DEFAULT_ROLE = RETAIL_DATA_ENGINEER_ROLE
20 DEFAULT_WAREHOUSE = RETAIL_LOAD_WH
21 MUST_CHANGE_PASSWORD = FALSE;
22
23 GRANT USAGE ON WAREHOUSE RETAIL_LOAD_WH TO ROLE RETAIL_DATA_ENGINEER_ROLE;
24 GRANT USAGE ON DATABASE RETAIL_DB TO ROLE RETAIL_DATA_ENGINEER_ROLE;
25 GRANT USAGE ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
26
27 GRANT ALL PRIVILEGES ON SCHEMA RETAIL_DB.ORDERS_DEMO TO ROLE RETAIL_DATA_ENGINEER_ROLE;
28
29 GRANT ROLE RETAIL_DATA_ENGINEER_ROLE TO USER RETAIL_ENGINEER_USER;
```

Results (just now)

	status
1	Statement executed successfully.

Part B: Azure Storage Setup

- 7. Create an Azure Storage Account.
- 8. Create a container named retail-data.
- 9. Inside the container, create this folder path:

raw/orders/

- 10. Upload the initial CSV file:

retail_orders_initial.csv

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

azuser5803_mm1.local@...
DEFAULT DIRECTORY (KARTHIKUR...

Home > retaildatastorage1 | Containers

retail-data

Container

Search

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔑 Acquire lease

🔑 Break lease

🔧 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

retail-data > raw > orders

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

	Name	Last modified	Access tier	Blob type	Size	Lease state
	📁 [.]					...
	📄 retail_orders_initial.csv	5/21/2026, 3:27:27 PM	Hot (Inferred)	Block blob	201 B	Available

Part C: Snowflake External Stage

11. Create a file format for CSV.

```
31
32 CREATE OR REPLACE FILE FORMAT RETAIL_DB.ORDERS_DEMO.CSV_FORMAT
33 TYPE = CSV
34 FIELD_DELIMITER = ','
35 SKIP_HEADER = 1
36 FIELD_OPTIONALLY_ENCLOSED_BY = '';
```

Results (just now)

Table Chart Pivot

1 row 106ms

#	status
1	File format CSV_FORMAT successfully created.

12. Create an Azure storage integration or external stage.

sv=2026-02-06&ss=b&srt=sco&sp=rlx&se=2026-05-21T17:59:56Z&st=2026-05-21T09:44:56Z&spr=https&sig=GT0yykXK%2FubW2izOisEeggQNqp8%2FNIGd2ejGvizwhWo%3D

13. Create an external stage pointing to:

azure://<storage_account>.blob.core.windows.net/retail-data/raw/orders/

14. List files from the stage.

```
My Workspace > codingassignment.sql
```

ACCOUNTADMIN RETAIL_LOAD_WH (X-Small) RETAIL_DB ORDERS_DEMO Share ...

```
29 GRANT ROLE RETAIL_DATA_ENGINEER_ROLE TO USER RETAIL_ENGINEER_USER;
30
31
32 CREATE OR REPLACE FILE FORMAT RETAIL_DB.ORDERS_DEMO.CSV_FORMAT
33 TYPE = CSV
34 FIELD_DELIMITER = ','
35 SKIP_HEADER = 1
36 FIELD_OPTIONALLY_ENCLOSED_BY = '';
37
38 CREATE OR REPLACE STAGE RETAIL_DB.ORDERS_DEMO.AZURE_STAGE
39 URL = 'azure://retaildatastorage1.blob.core.windows.net/retail-data/raw/orders/'
40 CREDENTIALS = (AZURE_SAS_TOKEN = 'sv=2026-02-06&ss=b&srt=sco&sp=rlx&se=2026-05-21T17:59:56Z&st=2026-05-21T09:44:56Z&spr=https&sig=GT0yykXK%2FubW2izOisEeggQNqp8%2FNIGd2ejGvizwhWo%3D')
41 FILE_FORMAT = RETAIL_DB.ORDERS_DEMO.CSV_FORMAT;
42
43 LIST @RETAIL_DB.ORDERS_DEMO.AZURE_STAGE;
```

Results (just now)

Table Chart Pivot

1 row 142ms

#	name	size	md5	last_modified
1	azure://retaildatastorage1.blob.core.windows.net/retail-data/raw/orders/retail_orders_initial.csv	201	749dbf85735430f758639efe29e6411	Thu, 21 May 2026 09:57:27 GMT

Part D: Initial Data Load

15. Create a raw landing table named ORDERS_RAW.

16. Create a final table named ORDERS_FINAL.

```
44
45 CREATE OR REPLACE TABLE ORDERS_RAW (
46     order_id INT,
47     product STRING,
48     amount INT,
49     price FLOAT
50 );
51
52 CREATE OR REPLACE TABLE ORDERS_FINAL (
53     order_id INT,
54     product STRING,
55     amount INT,
56     price FLOAT
57 );
58
59 SHOW TABLES;
60
```

Results (just now)

Table Chart Pivot 2 rows 45ms

	created_on	name	database_name	schema_name	kind	comment	cluster_by	# rows	# bytes	owner
1	2026-05-21 03:10:50.649 -0700	ORDERS_FINAL	RETAIL_DB	ORDERS_DEMO	TABLE			0	0	ACCOUNTADMIN
2	2026-05-21 03:10:49.799 -0700	ORDERS_RAW	RETAIL_DB	ORDERS_DEMO	TABLE			0	0	ACCOUNTADMIN

17. Load the initial file into ORDERS_RAW.

```
COPY INTO ORDERS_RAW
FROM @RETAIL_DB.ORDERS_DEMO.AZURE_STAGE
FILE_FORMAT = (FORMAT_NAME = CSV_FORMAT)
ON_ERROR = 'CONTINUE';
```

SELECT * FROM ORDERS_RAW;

Add to Chat Explain Quick edit Format

Results (just now)

Table Chart Pivot 10 rows 597ms

# ORDER_ID	PRODUCT	AMOUNT	PRICE
1	Camera	10.0%	50000
2	Charger	10.0%	800
3	Laptop	1	50000
4	Mouse	2	500
5	Keyboard	1	1500
6	Monitor	1	10000
7	Headphones	3	2000
8	Tablet	1	15000
9	Charger	2	800
10	Camera	1	25000

18. Insert data into ORDERS_FINAL.

```
67
68 INSERT INTO ORDERS_FINAL
69 SELECT * FROM ORDERS_RAW;
70
71 SELECT * FROM ORDERS_FINAL;
```

Results (just now)

Table Chart Pivot

10 rows 130ms

#	ORDER_ID	PRODUCT	AMOUNT	PRICE
1	1	Laptop	1	5000
2	2	Mouse	2	500
3	3	Keyboard	1	1500
4	4	Monitor	1	10000
5	5	Headphones	3	2000
6	6	Tablet	1	15000
7	7	Charger	2	800
8	8	Camera	1	25000

Part E: Incremental Loading

19. Create a Snowpipe object to load new files automatically into ORDERS_RAW.

```
72
73 CREATE OR REPLACE PIPE ORDERS_PIPE
74 AUTO_INGEST = FALSE
75 AS
76 COPY INTO ORDERS_RAW
77 FROM @RETAIL_DB.ORDERS_DEMO.AZURE_STAGE
78 FILE_FORMAT = (FORMAT_NAME = RETAIL_DB.ORDERS_DEMO.CSV_FORMAT);
```

Results (just now)

Table Chart Pivot

1 row 132ms

#	status
1	Pipe ORDERS_PIPE successfully created.

20. Create a stream on ORDERS_RAW.

The screenshot shows a SQL IDE interface with a workspace named 'codingassignment.sql'. The SQL editor contains the following code:

```
77 -- Ctrl+I to generate
80 CREATE OR REPLACE STREAM ORDERS_STREAM
81 ON TABLE ORDERS_RAW;
82
83 SELECT * FROM ORDERS_STREAM;
```

Below the editor, the 'Results (just now)' section is visible, showing a table with the following columns: ORDER_ID, PRODUCT, AMOUNT, PRICE, METADATA\$ACTION, METADATA\$ISUPDATE, and METADATA\$ROW_ID. The table is currently empty, and a warning icon is displayed at the bottom.

21. Create a task to merge new rows from the stream into ORDERS_FINAL.

The screenshot shows a SQL IDE interface with a workspace named 'codingassignment.sql'. The SQL editor contains the following code:

```
85
86 CREATE OR REPLACE TASK ORDERS_TASK
87 WAREHOUSE = RETAIL_LOAD_WH
88 SCHEDULE = '1 MINUTE'
89 AS
90 MERGE INTO ORDERS_FINAL T
91 USING ORDERS_STREAM S
92 ON T.ID = S.ID
93 WHEN MATCHED THEN UPDATE SET
94     T.PRODUCT = S.PRODUCT,
95     T.PRICE = S.PRICE,
96     T.AMOUNT = S.AMOUNT,
97     T.STORE_ID = S.STORE_ID
98 WHEN NOT MATCHED THEN INSERT
99 VALUES (S.ID, S.PRODUCT, S.PRICE, S.AMOUNT, S.STORE_ID);
```

Below the editor, the 'Results (just now)' section is visible, showing a table with the following columns: A_status. The table contains one row with the value 'Task ORDERS_TASK successfully created.'

22. Upload another CSV file into Azure Storage.

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo and a search bar. Below the navigation bar, the breadcrumb path is "Home > retaildatastorage1 | Containers". The main heading is "retail-data" with a subheading "Container". On the left, there's a sidebar with "Overview", "Diagnose and solve problems", "Access Control (IAM)", and "Settings". The "Overview" tab is selected. The main content area shows the "retail-data" container with a sub-path "raw > orders". There's a notification at the top right: "Successfully uploaded blob(s). Successfully uploaded 1 blob(s)." Below the notification, there's a search bar and a dropdown menu "Only show active blobs". The table shows "Showing all 2 items" with columns: Name, Last modified, Access tier, Blob type, Size, and Lease state. The items are "retail_orders_incremental_20260508.csv" and "retail_orders_initial.csv".

Name	Last modified	Access tier	Blob type	Size	Lease state
retail_orders_incremental_20260508.csv	5/21/2026, 3:53:26 PM	Hot (Inferred)	Block blob	115 B	Available
retail_orders_initial.csv	5/21/2026, 3:27:27 PM	Hot (Inferred)	Block blob	201 B	Available

23. Verify that new rows are automatically loaded into Snowflake.

The screenshot shows the Snowflake SQL editor interface. The workspace is "codingassignment.sql". The SQL command being executed is "ALTER PIPE ORDERS_PIPE REFRESH;". The results show "Results (just now)" with a table containing one row: "retail_orders_incremental_20260508.csv" with a status of "SENT".

File	Status
retail_orders_incremental_20260508.csv	SENT

The screenshot shows the Snowflake SQL editor interface. The workspace is "codingassignment.sql". The SQL command being executed is "SELECT * FROM ORDERS_STREAM;". The results show "Results (just now)" with a table containing 5 rows. The columns are: ORDER_ID, PRODUCT, AMOUNT, PRICE, METADATA\$ACTION, METADATA\$ISUPDATE, and METADATA\$ROW_ID.

ORDER_ID	PRODUCT	AMOUNT	PRICE	METADATA\$ACTION	METADATA\$ISUPDATE	METADATA\$ROW_ID
11	Item11	1	1000	INSERT	FALSE	89a485f1d7afa1aa021c67db768ebe7304a7447a
12	Item12	2	2000	INSERT	FALSE	2dfcc3dffd45abd707ccee08514939039770fed9
13	Item13	1	1500	INSERT	FALSE	f3d440b8a9432335545e48208fc20efb8fe8885a
14	Item14	3	3000	INSERT	FALSE	78e03a5f0f77e189cbaf8e715f57a94adaebcd67
15	Item15	2	2500	INSERT	FALSE	826c5f6c96c017d4f651875441c05428f562eb83

codingassignment.sql

My Workspace > codingassignment.sql

ACCOUNTADMIN RETAIL_LOAD_WH (X-Small) RETAIL_DB ORDERS_DEMO Share

102
103 SELECT * FROM ORDERS_RAW;
104
105

Results (just now)

Table Chart Pivot

15 rows 154ms

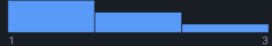
ORDER_ID



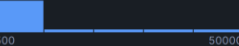
PRODUCT

Camera
Charger
+13 more

AMOUNT



PRICE



1	1	Laptop	1	50000
2	2	Mouse	2	500
3	3	Keyboard	1	1500
4	4	Monitor	1	10000
5	5	Headphones	3	2000
6	6	Tablet	1	15000
7	7	Charger	2	500
8	8	Camera	1	2500
9	9	Tripod	1	3000
10	10	Speaker	2	4000
11	11	Item11	1	1000
12	12	Item12	2	2000
13	13	Item13	1	1500
14	14	Item14	3	3000
15	15	Item15	2	2500

A thin, solid horizontal line spanning the width of the page.